USB PD CTS ENGINEERING CHANGE NOTICE

NOTICE: Any Company or Companies submitting a USB Power Delivery ECR proposal must be one of the following: a Promoter or Contributor of the USB 3.0 and 2.0 Specifications who have completed the USB Power Delivery addendum. If a group of Companies is submitting an ECR proposal, each company must be either a Promoter or Contributor of the USB 3.0 and 2.0 Specifications who have completed the USB Power Delivery addendum.

SPECIFICATION REVISIONS AND ADDENDA: At any point in time, there shall only be one current version of the USB PD CTS, termed the production version. At the same time, there may also be proposed revisions to the specification's design which are not yet approved and shall be held confidential as deemed necessary by the USB 3.0 and USB 2.0 Promoters and within the Group of Working Committee(s).

PROCEDURES FOR SUBMITTING PROPOSALS: Both members of the USB Implementers Forum as a whole and members of the USB 3.0 and USB 2.0 Promoters may submit requests to revise the USB PD CTS Specification. Such a request may be rejected or may result in a USB PD Engineering Change Notice (ECN), which is the official way USB specifications may be changed.

FORMAT OF PROPOSAL: The originator of a request to alter the USB PD CTS Specification may do so by posting this to the USB Power Delivery Compliance working group for review. Once the proposal has been reviewed by the working group it will be passed to the USB 3.0 and 2.0 Promoters for approval to publish.

RESUBMISSION AND APPEAL: The originator of a request that was not approved can redraft the original request. Rewritten proposal will be treated as a new proposal and will be evaluated using the procedures described above. The originator of a request that was not approved can also submit an appeal to the USB 3.0 and 2.0 Promoters. The appeal must be made in writing and addressed to the Secretary of the USB Implementers Forum.

ABOUT THE ENGINEERING CHANGE REQUEST FORM:
The Purpose of this Engineering Change Request Form is to expedite the review process of the proposal by providing explanations, background information, and examples of the proposed changes at a high level. This form serves as an executive summary to the actual proposal.

STEPS ON HOW TO SUBMIT A USB PD ENGINEERING CHANGE REQUEST:

a. Please fill out the Engineering Change Request Form on the following pages completely:
    a. Detail the names and contact details for each of the ECR contributors
    b. Update the ECR Title
    c. Give a minimum of 2-3 sentences for each description on the form outlining the background to the ECR
b. For each section/table/figure to be updated:
    a. Detail the section number, starting page and figure/table number to be updated as appropriate.
    b. Detail existing text under "From Text"
    c. Detail changed text under "To Text"
c. Save the file as "USB PD CTS 1.0 R 1" followed by the ECR Title as per step a.b
d. Post the ECR in the USB PD CTS Documents section under "ECR | New ECRs".
    a. This ECR will then be reviewed by the Power Delivery Compliance Working Group.
    b. Revisions to the ECR originating from the review should be submitted as document revision of the original ECR using "Add new document".

USB PD CTS ENGINEERING CHANGE NOTICE

Name: Ballot Nathalie Email: Nathalie.ballot@st.com

Company: STMicroelectronics Mailstop:

Address:

City: State/Province:

Country: Zip/Postal Code:

Phone: FAX:

Name: Email:

Company: Mailstop:

Address:

City: State/Province:

Country: Zip/Postal Code:

Phone: FAX:

Name: Email:

Company: Mailstop:

Address:

City: State/Province:

Country: Zip/Postal Code:

Phone: FAX:

Name: Email:

Company: Mailstop:

Address:

City: State/Province:

Country: Zip/Postal Code:

Phone: FAX:

Name: Email:

Company: Mailstop:

Address:

City: State/Province:

Country: Zip/Postal Code:

Phone: FAX:

USB PD CTS ENGINEERING CHANGE NOTICE

Title: TEST.PD.PS.SCR.1 :Multiple Request Load Test
Applied to: USB PD CTS Specification Q3 2026

Brief description of the functional changes proposed:
Some Sources reject EPR entry if EPR_capable has been set after the 1st explicit contract but prior to receive EPR_enter mode

Benefits as a result of the proposed changes:
Alignment to main specification

An assessment of the impact to the existing revision and systems that currently conform to the USB specification:
None if DUT properly implemented

An analysis of the hardware implications:
none

An analysis of the software implications:
none

An analysis of the compliance testing implications:
Tester SW shall be updated

An analysis of the Vendor Info File (VIF) implications:
None

USB PD CTS ENGINEERING CHANGE NOTICE

Actual Change Requested

(a). Section x.x.x, Page x, Figure/Table x-x

FROM:

TEST.PD.SRC3.1 *EPR Entry Process – UUT as VCONN Source*

Description: The Tester as a sink verifies the successful EPR negotiation when UUT acts as a VCONN source.

Test Specific Tester Behavior:

Test Conditions:

	Consumer Only	Provider Only	DRP, C/P, P/C
Rev3ChkdSrc		✓	✓
Rev3UnChkdSrc		✓	✓

Test Procedures:

1. There are 2 possible bring-up procedures:

    a. The UUT has VIF field PD_Port_Type set to Provider Only, Provider/Consumer or DRP, the Tester behaves as a Sink Only and it runs bring-up procedure with the UUT as a Source COMMON.PROC.BU.1

    b. The UUT has VIF field PD_Port_Type set to Consumer/Provider, the Tester behaves as a Provider/Consumer, and it runs bring-up procedure with the UUT as a Source COMMON.PROC.BU.7. If the UUT does not accept V_{CONN}_Swap to become V_{CONN} Source, the test passes and stops here.

2. The Tester sends the *EPR_Get_Source_Cap* Message and perform below check [TEST.PD.EPR.SRC3.1#13]

    a. If the UUT is Provider Only and VIF field EPR_Supported_as_SRC is No, the tester checks that the UUT responds with the *Not_Supported* Message.

    b. If the UUT is DRP or C/P or P/C and VIF field EPR_Supported_as_SNK is No and VIF field EPR_Supported_as_SRC = No , the Tester checks that the UUT responds with the *Not_Supported* Message.

    c. If the UUT is not Provider Only and VIF field EPR_Supported_as_SRC is No and VIF field EPR_Supported_as_Snk is Yes, the Tester checks that the UUT responds the *Reject* or *Not_Supported* Message.

    d. If the UUT VIF field EPR_Supported_as_SRC is Yes, the Tester checks that the UUT responds with the *EPR_Source_Capabilities* Message.

Note: *EPR_Source_Capabilities* should not be the maximum power the UUT is able to offer.

3. The Tester sends an *EPR_Mode (* Enter) Message with the Action field set to 0x01(Enter) and Data field set to 140 (140W)

To:

TEST.PD.SRC3.1 *EPR Entry Process – UUT as VCONN Source*

Description: The Tester as a sink verifies the successful EPR negotiation when UUT acts as a VCONN source.

Test Specific Tester Behavior:

Test Conditions:

	Consumer Only	Provider Only	DRP, C/P, P/C
Rev3ChkdSrc		✓	✓
Rev3UnChkdSrc		✓	✓

Test Procedures:

1. There are 2 possible bring-up procedures:

   a. The UUT has VIF field PD_Port_Type set to Provider Only, Provider/Consumer or DRP, the Tester behaves as a Sink Only and it runs bring-up procedure with the UUT as a Source COMMON.PROC.BU.1 Request is sent with EPR_capable (bit22) = 0

   b. The UUT has VIF field PD_Port_Type set to Consumer/Provider, the Tester behaves as a Provider/Consumer, and it runs bring-up procedure with the UUT as a Source COMMON.PROC.BU.7. If the UUT does not accept V_{CONN}_Swap to become V_{CONN} Source, the test passes and stops here. Request is sent with EPR_capable (bit22) = 0

   c. If EPR_Supported_as_SRC is Yes the tester re-iterate the last Request message with EPR_capable bit (bit22) = 1. DUT shall proceed by sending accept and PSRdy message.

2. The Tester sends the *EPR_Get_Source_Cap* Message and perform below check [TEST.PD.EPR.SRC3.1#13]

   a. If the UUT is Provider Only and VIF field EPR_Supported_as_SRC is No, the tester checks that the UUT responds with the *Not_Supported* Message.

   b. If the UUT is DRP or C/P or P/C and VIF field EPR_Supported_as_SNK is No and VIF field EPR_Supported_as_SRC = No , the Tester checks that the UUT responds with the *Not_Supported* Message.

c. If the UUT is not Provider Only and VIF field EPR_Supported_as_SRC is No and VIF field EPR_Supported_as_Snk is Yes, the Tester checks that the UUT responds the *Reject* or *Not_Supported* Message.

d. If the UUT VIF field EPR_Supported_as_SRC is Yes, the Tester checks that the UUT responds with the *EPR_Source_Capabilities* Message.

Note: *EPR_Source_Capabilities* should not be the maximum power the UUT is able to offer.

3. The Tester sends an *EPR_Mode (* Enter) Message with the Action field set to 0x01(Enter) and Data field set to 140 (140W)